

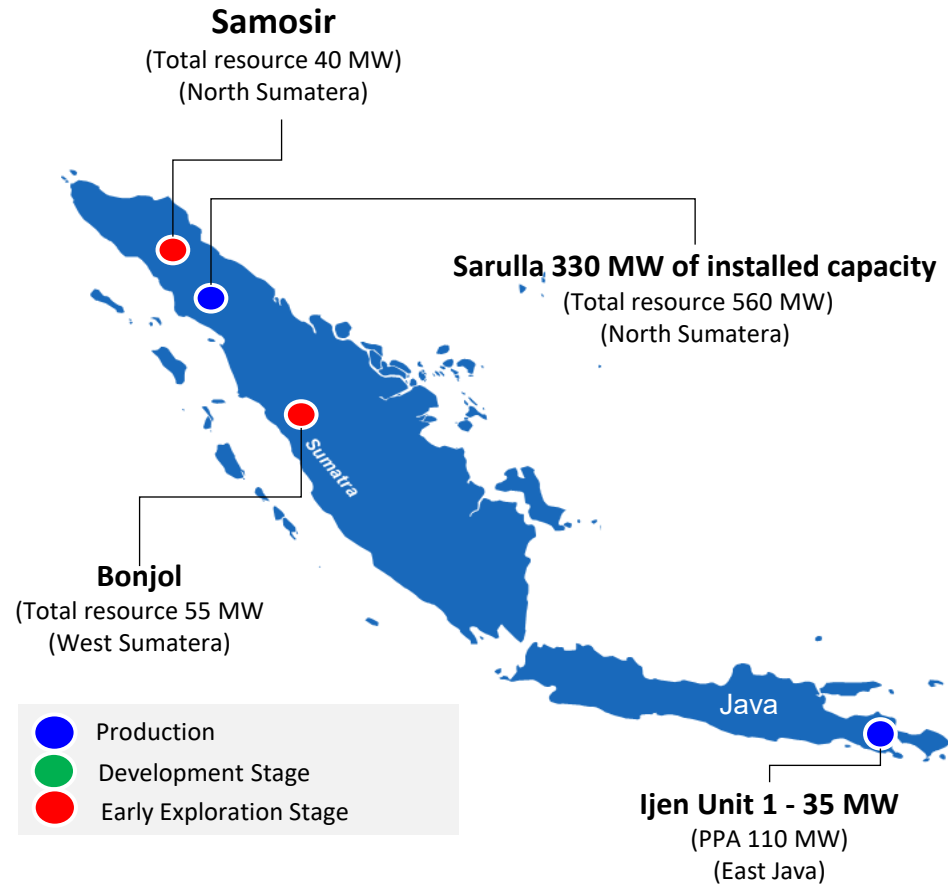
BLAWAN IJEN GEOTHERMAL POWER PLANT UNIT 1 (35 MW) LARGEST TWO-PHASE BINARY SYSTEM IN INDONESIA

PT Medco Cahaya Geothermal
Bondowoso, August 2025



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Geothermal

MPI Portfolio on Operating and Developing Geothermal Projects



- ❑ Sarulla → started operation since 2018
- ❑ Ijen → started operation since feb 2025
- ❑ Bonjol → execute exploration drilling on Agustus 2025
- ❑ Samosir → initiates the geoscience survey in 2025



Samosir, Numerous Fumaroles, Steam Vents and Hot springs, Resource potency 40 MW



Ijen, Geothermal Project Ijen of 35 MW unit 1, Total resource capacity 110 MWe



Sarulla, 330 MW installed capacity in Sarulla geothermal operation



Bonjol, Numerous Hot Springs, exploration of resource potency 60 MW

MEDCO POWER INDONESIA

Focus on Pipeline Projects to Target
Total of 500 MW Developments



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IJEN GEOTHERMAL POWER PLANT SUMMARY



Blawan Ijen Geothermal Power Plant Unit 1

The first geothermal power plant in East Java with a capacity of 35 MW, the plant is capable of supplying enough green energy to light up at least 75,000 households. The existence of the PLTP not only supports the stability of the eastern electricity system but also contributes significantly to the clean energy transition, sustainable geothermal utilization and local economy improvement

Net Capacity	: 35 MW (URC)
COD	: 09 February 2025
Plant Location	: Bondowoso, East Java
PPA Contract	: 110 MW



Well information:

Active for Power Plant:

- 6 Production Wells
- 1 Injection Well



Our People:

- Exploration phase: 278 employees
- Development phase: 1,010 employees
- Operation phase: 86 employees



Power Transmission:

28 km for 150 kV Transmission Line (83 towers)
from Power Plant to Banyuwangi Substation



Achievements / Award:

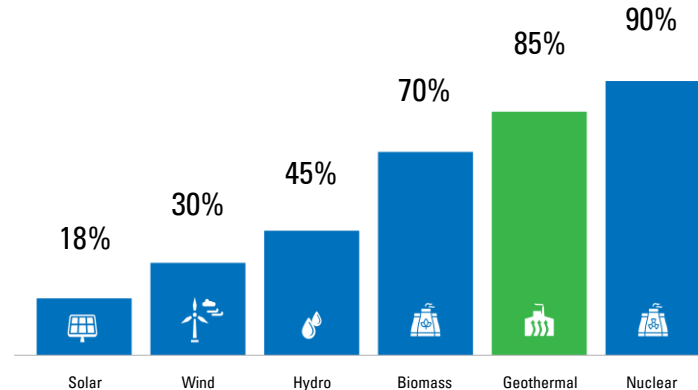
- ✓ IBEA Award 2025 – Best IPP in Technology Innovation for Renewable Energy
- ✓ K3LL Award by East Java Governor (Jan 2025)
- ✓ Subroto Award by EBTKE (Dec 2024)
- ✓ Zero Accident Award by Ministry of Manpower (Jan 2024)
- ✓ ESG Award by PT SMI (Sept 2024)
- ✓ HIV-AIDS and Tuberculosis Prevention Award by East Java Government (2024)
- ✓ Subroto Award by EBTKE (2023)

IJEN GPP Inauguration by Mr. President & MEMR

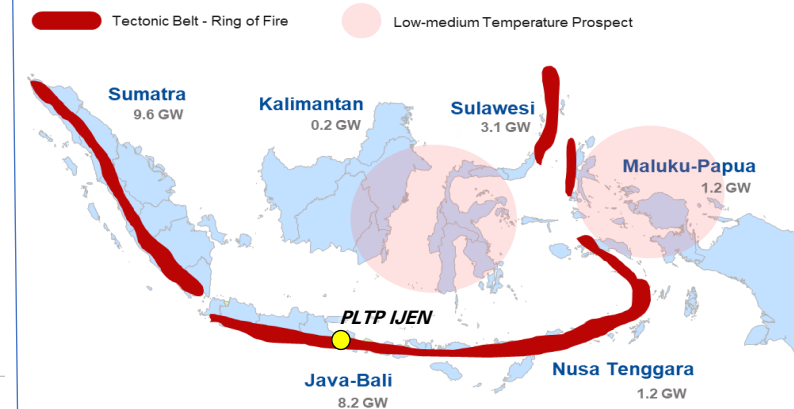


- The President and MEMR expressed strong support for geothermal energy, in line with the SWASEMBADA ENERGI (Energy Self-Sufficiency) concept, which aligns with Indonesia's high geothermal reserves and capacity factors (CF)
- PLTP Ijen is well-aligned with this vision, having consistently achieved 90% TOP during the first six months of early production
- Medco Cahaya Geothermal is committed to continuing excellent operations and further expanding the plant's capacity up to 110 MW under the Power Purchase Agreement (PPA)
- Binary power plants represent cutting-edge technology, offering enhanced flexibility in utilizing both steam and brine resources

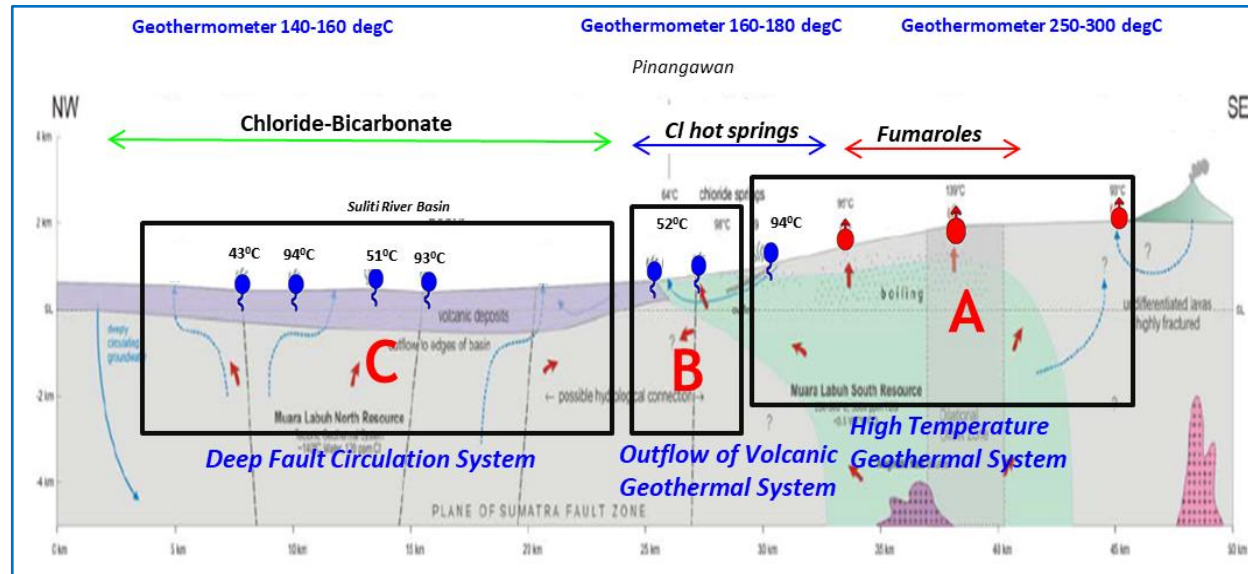
Baseload Electricity with High CF



Indonesia Geothermal Resources



POWER PLANT TECHNOLOGY – TWO PHASE BINARY



Remark: Type A : High Temperature System (well-developed in Indonesia)
Type B and C : Intermediate Temp System (Future of Indonesia)

- Medco has collaborative action with Ormat Technology (Power Plant) → Bring the technology into Indonesia
- Ormat has 83% of world binary power plant market, or 18% of total power plant market (Wolf, 2021)
- Intermediate temperature system → Future geothermal of Indonesia → potential binary system
- Step by step geothermal development, collaborate with National & International partner can increase success ratio of Project

Exploration

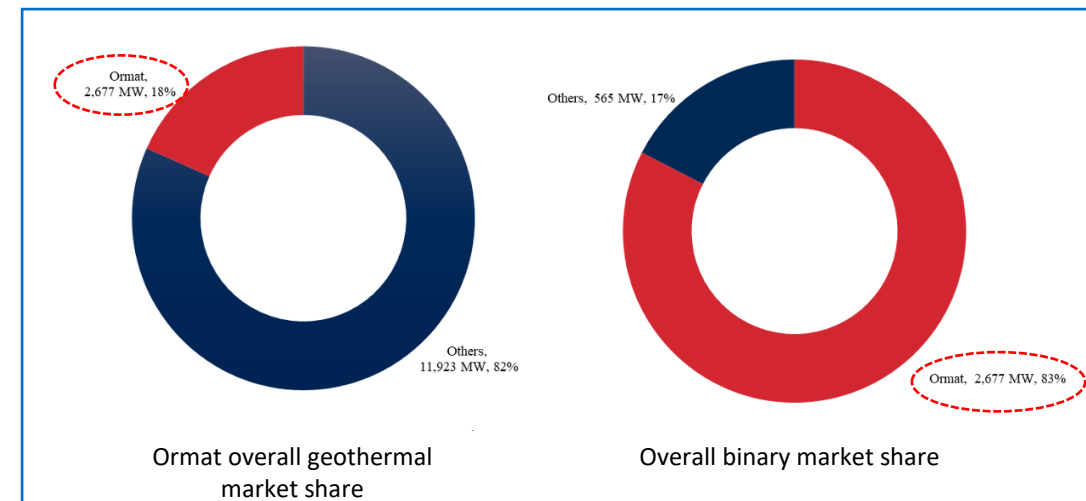
- Drilling Rig : EPI
- Well Logging : SDI & Halliburton
- Geoscience : Jacobs, GTX, ELC, UI

Dev & Construction

- Power Plant : Ormat
- Construction : IKPT
- Construction : Multifab

Production & Monitoring

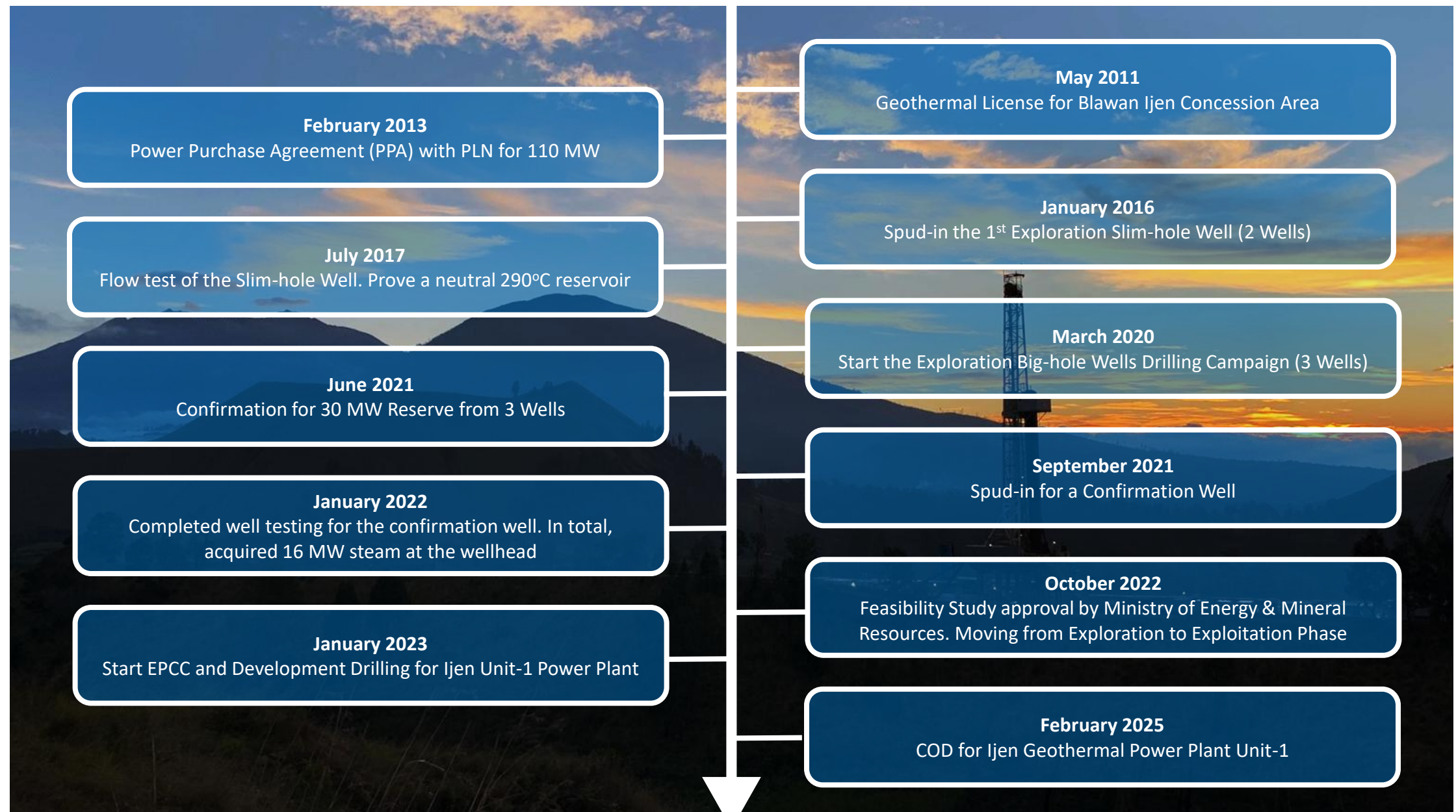
- MEQ : ITB
- Chemistry Monitoring : Thermochem
- Microgravity : Elnusa



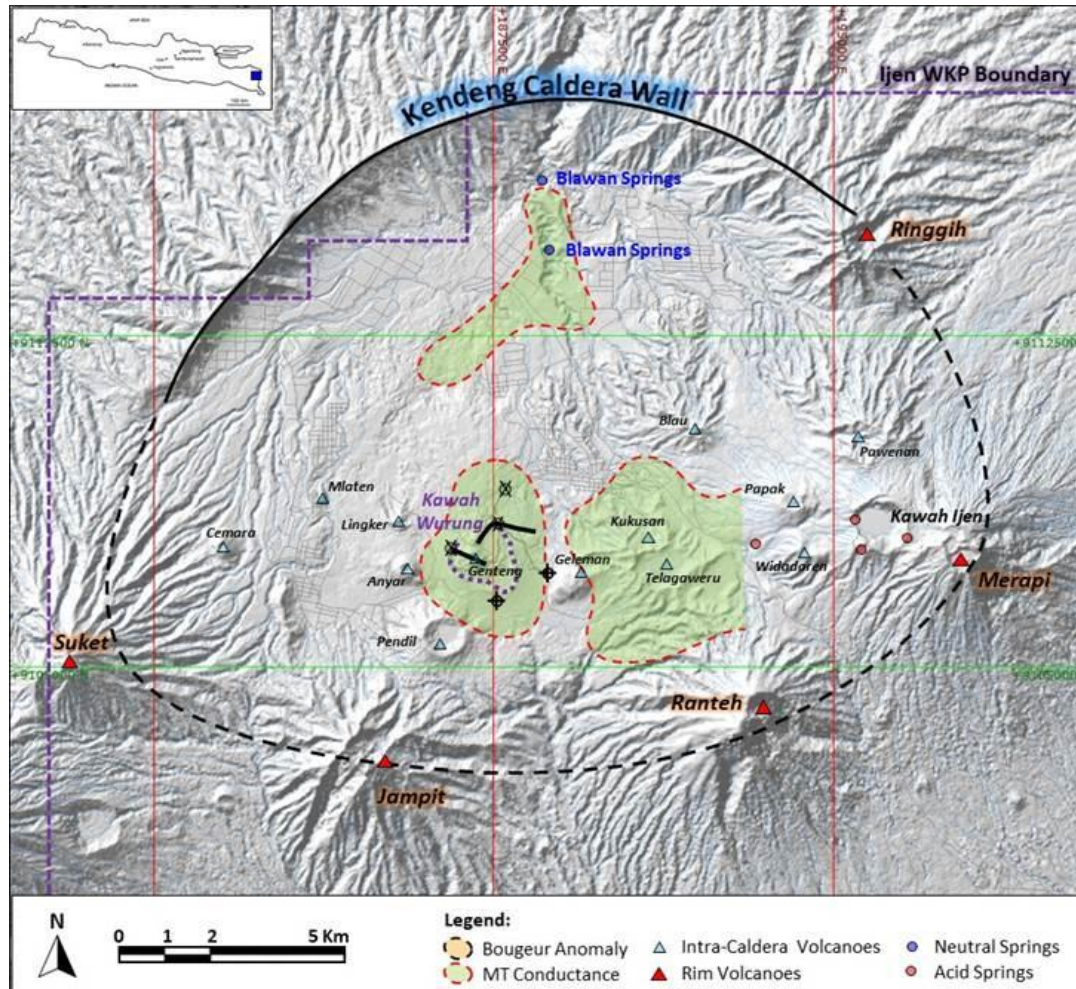


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PROJECT HISTORY



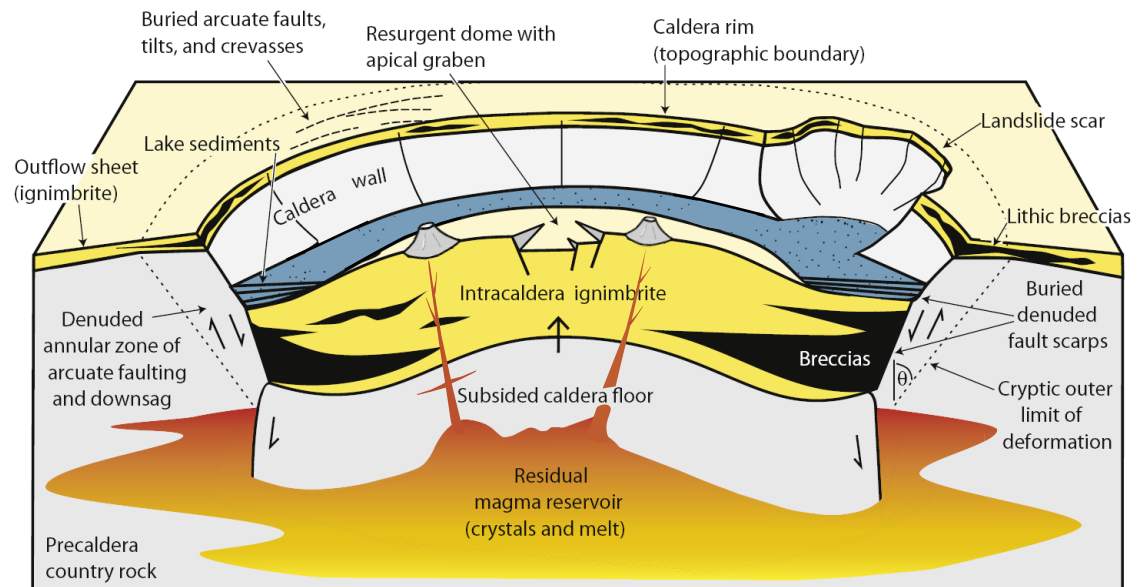
IJEN GEOTHERMAL PROSPECT – KENDENG CALDERA



3 prospects in Kendeng Caldera:

- 1) **Kawah Wurung developed 35 MW (with possible additional 20-25 MW)**
- 2) **Telaga Weru prospect possible 45-50 MW**
- 3) **Blawan prospect intermediate system**

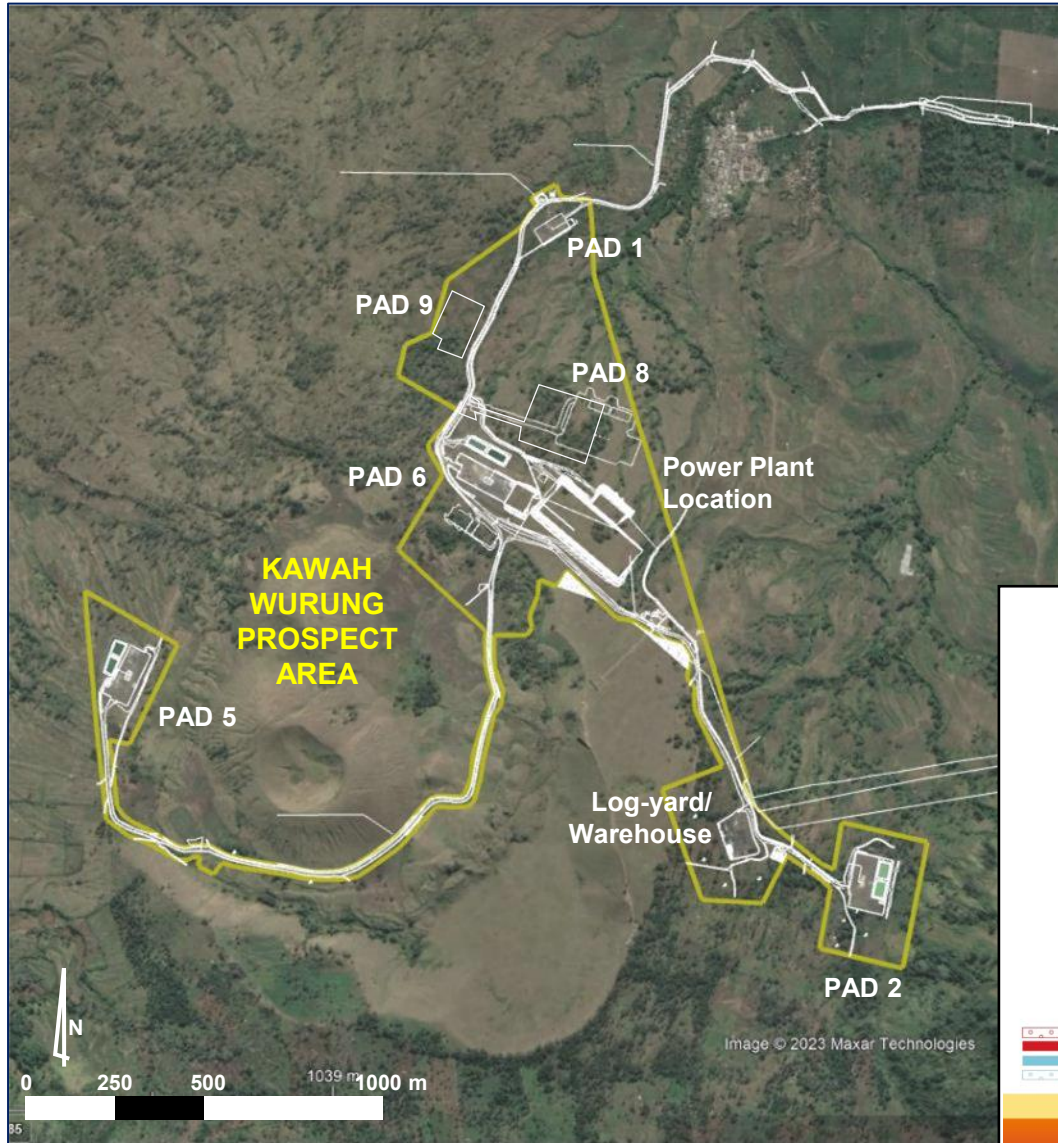
- Geothermal Systems associated with heat source from caldera dome, intra-caldera volcanoes and rim volcanoes
- Manifestation limited at Kawah Ijen (acid springs) and Blawan (neutral thermal springs)
- One of the prospects is characterized as a “blind geothermal system”. Courageous and conservative exploration conducted, then, in February 2025, the first geothermal power plant of Ijen Unit-1 started the operation. The next units planned to be developed, utilizing a similar approach.
- Exploration drilling confirmed the existence of neutral geothermal reservoir, a thin steam cap on top of liquid dominated reservoir with **temperature up to 325°C**.





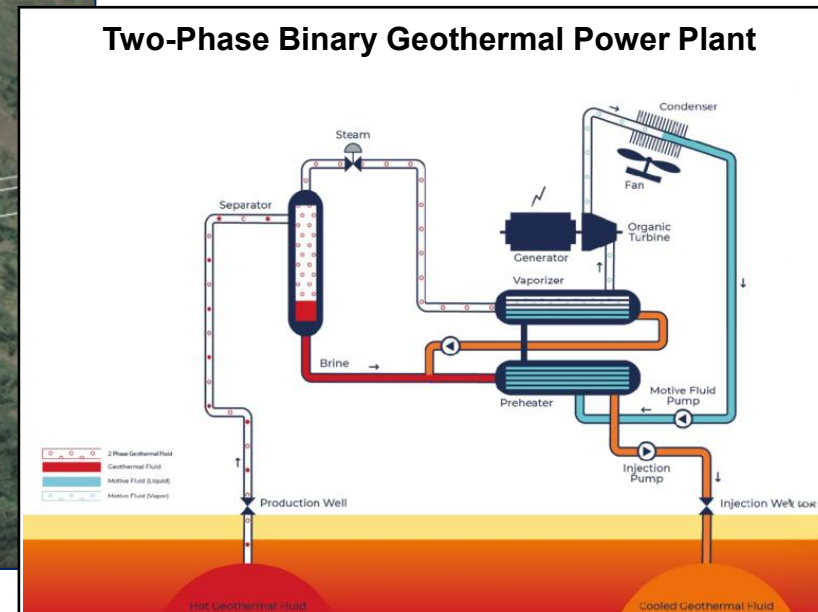
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Technology Selection : Two-Phase Binary System



Power Plant and Transmission line construction

- ✓ Numerical model confirms the sustainability of 35 MW for project life
- ✓ Exploration-Development drilling produces wells with enthalpy 1350 – 1600 kJ/kg, **technology selection** → **full Two phase-binary system**
 - ✓ TMF ~140 kg/s
 - ✓ Steam ~60 kg/s
 - ✓ Brine ~80 kg/s
- ✓ Benefit of Two-Phase binary system:
 - ✓ 100% injection → optimum pressure support to reservoir (the monitoring is on progress)
 - ✓ Pressure support → Reduce make up well (on progress monitoring)
 - ✓ Higher generation (Preheater & Vaporizer)



Two-phase binary system schematic



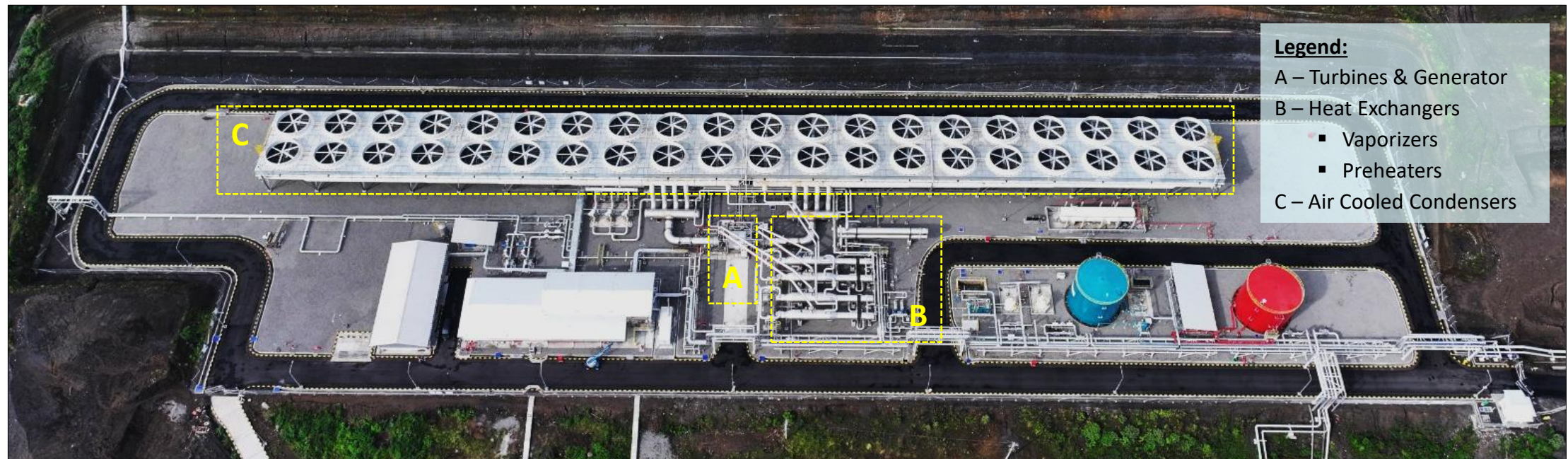
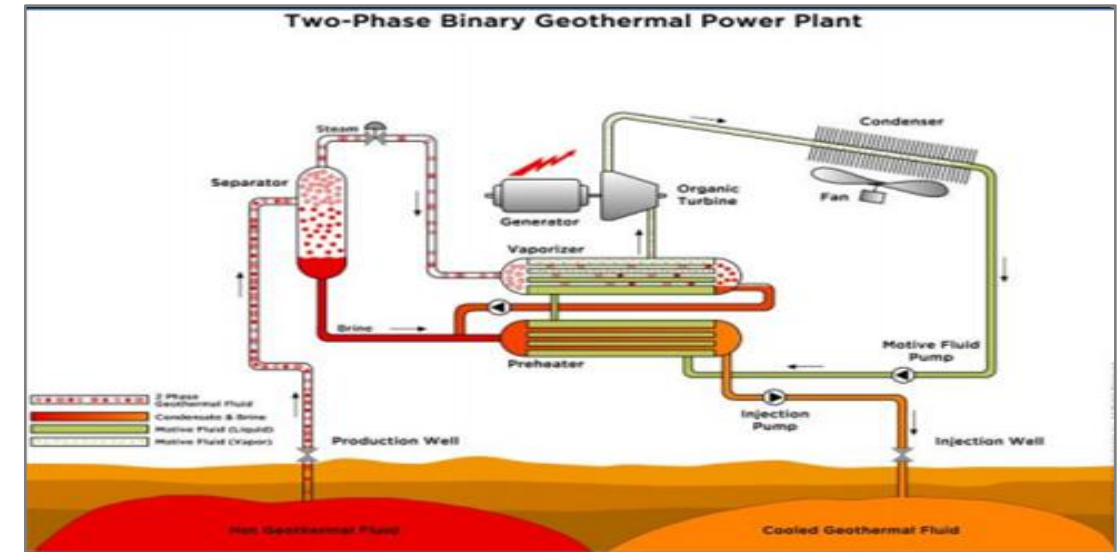
150 kV Transmission line

POWER PLANT TECHNOLOGY – TWO PHASE BINARY

Two-phase binary powerplant (OEC) is utilizing steam and brine from production wells to heat-up motive fluids, which then rotate turbines (2x17.5 MW) and generate the electricity.

Steam will go to vaporizers (4 units). Then, produced steam condensate mixed with brine and go to preheaters (2 units). Afterward, the mixed fluids will be reinjected to subsurface via injection well. After motive fluids (cyclopentane) rotate the turbine, it will be cooled down inside the ACC, before returning back to the OEC.

This system is 100% reinjection, which means more environmentally friendly and better to maintain the reservoir pressure (more sustain production).



SUSTAINABILITY AND COLLATERAL IMPACT (LOCAL COMMUNITY)

Medco Pintar



Medco Lestari



Medco Peduli



"We are partnering with stake holders and involving local community in every steps of our development"



65 Project/Activity

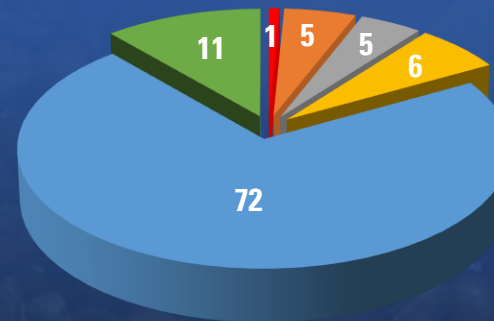


Expected beneficiaries:
7.090 Peoples/17 Groups/1400 Household
Around 9 sub-villages/10 Villages/ 8 sub-regency/2 regency



Top 3 dominant program :
Medco Peduli; Medco Membangun Negeri; Medco Berdaya

% Realization



■ Medco Pintar
■ Medco Lestari
■ Medco Peduli

■ Medco Sehat
■ Medco Berdaya
■ Medco Membangun Negeri

Medco Sehat



Medco Berdaya



Medco Membangun Negeri



HARMONY WITH LOCAL TOURISM

Blawan-Ijen Geothermal is located near several Ijen Geopark's Geosites, especially Kawah Wurung Geosite, which located adjacent to the power plant location. Other Geosites are Lava Flow Plalangan, Blawan Hotsprings & Lava, Gentongan Waterfall, Kalipait Acidic Stream, Ijen Acidic Crater Lake, and Ijen Blue-fire.

Kawah Wurung Crater:



Ijen Blue-fire:



Ijen Crater:



Kalipait Acidic Stream:



Gentongan Waterfall:



Blawan Hotsprings:



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Project Medco Cahaya Geothermal

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Macan, Kawah Wurung, Desa Kalianyar, Kecamatan
Ijen Kabupaten Bondowoso.

Website : www.medcopower.co.id

